

Normal Dist & Sampling

\$200

Scores on a standardized test are normally distributed with mean 500 and standard deviation 100. By the Empirical Rule, approximately this percentage of test-takers scored between 400 and 600.

Answer What is 68%?**Worked Solution**

$400 = 500 - 100 = \mu - 1 \text{ SD}$ and $600 = 500 + 100 = \mu + 1 \text{ SD}$. By the Empirical Rule, 68% of data falls within 1 standard deviation of the mean.

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A population has mean $\mu = 50$ and standard deviation $\sigma = 20$. For random samples of size $n = 25$, this is the standard error of the sampling distribution of $\bar{x}$.

Answer What is 4?**Worked Solution**

Standard error = $\sigma / \sqrt{(n)} = 20 / \sqrt{(25)} = 20 / 5 = 4$.

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A fair six-sided die is rolled 15 times. The mean of the first 10 rolls is $\bar{x}_1$ and the mean of the remaining 5 rolls is $\bar{x}_2$. The mean of the sampling distribution of $(\bar{x}_1 - \bar{x}_2)$ equals this value.

Answer What is 0?

Worked Solution

For a fair die, $\mu = 3.5$. Both groups have the same population mean, so $\mu(\bar{x}_1 - \bar{x}_2) = 3.5 - 3.5 = 0$.

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For the sampling distribution of the sample proportion $\hat{p}$ to be approximately Normal, both of these two conditions involving n and p must be satisfied.

Answer What is $np \geq 10$ and $n(1-p) \geq 10$ (the Large Counts condition)?

Worked Solution

Large Counts condition: $np \geq 10$ and $n(1-p) \geq 10$. When both are satisfied, the sampling distribution of $\hat{p}$ is approximately Normal with mean p and SD $= \sqrt{p(1-p)/n}$.

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A strongly right-skewed population has mean 100 and SD 30. This theorem guarantees that for large samples ($n \geq 30$), the sampling distribution of $\bar{x}$ is approximately Normal, regardless of the population shape.

Answer What is the Central Limit Theorem (CLT)?

Worked Solution

CLT: for random samples of size $n \geq 30$ from any population with mean μ and SD σ , the sampling distribution of $\bar{x}$ is approximately Normal with mean μ and standard error $\sigma / \sqrt{n}$.

2-Var Quantitative Data

\$200 A data set has: mean HS GPA = 3.45, mean College GPA = 3.10, $S_x = 0.40$, $S_y = 0.55$, $r = 0.68$. The slope of the LSRL for predicting college GPA from HS GPA equals this value.

Answer What is 0.935?

Worked Solution

$$b = r \times (S_y / S_x) = 0.68 \times (0.55 / 0.40) = 0.68 \times 1.375 = 0.935.$$

$$a = \bar{y} - b(\bar{x}) = 3.10 - 0.935(3.45) = -0.126. \text{ LSRL: } \hat{y} = -0.126 + 0.935x.$$

\$400 Using a regression model to make a prediction for an x-value that is far outside the range of observed data is called this.

Answer What is extrapolation?

Worked Solution

Extrapolation: using a model to predict y for x-values outside the range of the observed data. The relationship may not hold beyond the data range, making the prediction unreliable.

\$600 A single data point that has an extreme x-value and substantially changes the slope or intercept of the regression line when included is called this type of point.

Answer What is an influential point?

Worked Solution

An influential point has an extreme x-value and substantially changes the slope or intercept when included vs. excluded. Fit the LSRL with and without the point to check.

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When a scatterplot shows an exponential curved relationship, a common transformation applied to the y-variable to linearize the data before regression is this.

Answer What is a log transformation?

Worked Solution

Taking $\log(y)$ often straightens an exponential relationship. After transforming, check the residual plot of the new model for random scatter to confirm the transformation worked.

\$1000

A computer output shows $t = 3.8$ for the slope of the LSRL with $p\text{-value} = 0.003$. At $\alpha = 0.05$, this is the correct conclusion about the population slope β .

Answer What is reject H_0 : there is convincing evidence of a linear relationship between x and y?

Worked Solution

$H_0: \beta = 0$. $p\text{-value} = 0.003 < \alpha = 0.05$, so reject H_0 . There is convincing evidence that a linear relationship exists between the variables.

Inference: Proportions

\$200 In the U.S., 36% of people have blood type A positive. For a one-sample z-test using 150 Norwegians, the standard deviation used in the test statistic is this.

Answer What is $\sqrt{(0.36 \times 0.64 / 150)}$?

Worked Solution

For a significance test, use the null hypothesis value p_0 in the standard deviation: $\sqrt{(p_0(1-p_0)/n)} = \sqrt{(0.36 \times 0.64 / 150)} \approx 0.039$.

\$400 A 95% confidence interval for the proportion of voters supporting a candidate is (0.48, 0.56). This is the correct interpretation of this interval.

Answer What is: we are 95% confident the true proportion of all voters who support the candidate is between 0.48 and 0.56?

Worked Solution

We are 95% confident the true population proportion falls between 0.48 and 0.56. The interval estimates the parameter, not individual voters.

\$600 In a sample of 150 Norwegians, 66 had blood type A positive. Before performing inference, you verify that $66 \geq 10$ and $84 \geq 10$. This is the name of the condition you just checked.

Answer What is the Large Counts condition?

Worked Solution

Large Counts condition for a confidence interval: $n(\hat{p}) \geq 10$ and $n(1-\hat{p}) \geq 10$. Here: $66 \geq 10$ and $84 \geq 10$. Both pass, so the sampling distribution of $\hat{p}$ is approximately Normal.

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Students were randomly assigned to standard or adaptive practice. The 99% CI for the difference in passing proportions (standard - adaptive) is (0.03, 0.24) and the 90% CI is (0.06, 0.21). At $\alpha = 0.05$, this is the conclusion of the two-proportion z-test.

Answer **What is reject H_0 : convincing evidence that the passing proportions differ?**

Worked Solution

Both CIs exclude 0. Since the 99% CI excludes 0, we reject H_0 at $\alpha = 0.01$, and therefore at $\alpha = 0.05$ as well. There is convincing evidence that the passing proportions differ.

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Researchers test $H_0: p_D = p_P$ vs. $H_a: p_D > p_P$ at $\alpha = 0.01$ and get p-value = 0.12. If the drug actually works, the researcher's conclusion of 'no effect' constitutes this type of error.

Answer **What is a Type II error (failing to reject a false null hypothesis)?**

Worked Solution

$p = 0.12 > \alpha = 0.01$, so fail to reject H_0 . If the drug actually works, H_0 is false, and failing to reject it is a Type II Error.

Type I Error: rejecting a true H_0 . Type II Error: failing to reject a false H_0 .
Power = $1 - P(\text{Type II Error})$.

Inference: Means

\$200 A 95% confidence interval for the mean weight of African bush elephants is (12.2, 14.6) thousand pounds. This is the correct interpretation of this interval.

Answer **What is: we are 95% confident the true mean weight of all African bush elephants is between 12.2 and 14.6 thousand pounds?**

Worked Solution

We are 95% confident the true population mean weight falls between 12.2 and 14.6 thousand pounds. The interval is about the parameter, not individual elephants.

\$400 A one-sample t-test for means is conducted using a random sample of $n = 30$ students. The number of degrees of freedom used to find the p-value is this.

Answer **What is 29?**

Worked Solution

$df = n - 1 = 30 - 1 = 29$.

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68 volunteers are randomly split into two groups: one works standing, the other sitting. After 9 months, mean back problems are compared. The most appropriate inference procedure is this.

Answer What is a two-sample t-test for a difference in means?

Worked Solution

Two separate independent groups, each receiving one treatment. The appropriate test is a two-sample t-test for $\mu_1 - \mu_2$. This is not matched pairs because each subject only experiences one condition.

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A consumer group takes samples of 100 airplanes from each of two airlines and constructs a 95% CI for the difference in mean seat widths. If the sample size decreases to 50 per airline (all else equal), this is the effect on the confidence interval.

Answer What is the interval becomes wider?

Worked Solution

Smaller n increases the standard error, which increases the margin of error, making the interval wider. Decreasing n from 100 to 50 means less precision.

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A sociologist compares salaries of bachelor's ($\bar{x} = 55,000$, $s = 3,500$, $n = 32$) and master's ($\bar{x} = 58,000$, $s = 4,000$, $n = 28$) degree holders. With $H_0: \mu_M - \mu_B = 0$, this is the correct formula for the two-sample t-test statistic.

Answer What is $t = (58,000 - 55,000) / \sqrt{(4,000^2 / 28 + 3,500^2 / 32)}$?

Worked Solution

$$t = (\bar{x}_1 - \bar{x}_2) / \sqrt{(s_1^2/n_1 + s_2^2/n_2)} = (58,000 - 55,000) / \sqrt{(4000^2/28 + 3500^2/32)}.$$
$$= 3000 / \sqrt{(571,429 + 382,813)} = 3000 / \sqrt{(954,241)} \approx 3000 / 977 \approx 3.07.$$
Add variances (s^2/n), not standard deviations.

Chi-Square Testing

\$200 A chi-square test for homogeneity across four high schools comparing absentee rates gave a test statistic of 19.02 and p-value of 0.025. Assuming H_0 states all schools have the same rates, this is the correct interpretation of the p-value.

Answer What is: assuming absentee rates are the same for all four schools, there is a 2.5% probability of observing a test statistic of 19.02 or larger?

Worked Solution

The p-value assumes H_0 is true. If absentee rates are equal across schools, a test statistic of 19.02 or larger would occur only 2.5% of the time. Since $0.025 < 0.05$, reject H_0 .

\$400 In a chi-square test, the formula used to calculate each cell's contribution to the overall test statistic is this.

Answer What is $(\text{Observed} - \text{Expected})^2 / \text{Expected}$?

Worked Solution

χ^2 = sum of $(O - E)^2 / E$ over all cells. Expected count = (row total $\times$ column total) / grand total.

\$600 A chi-square test of independence is conducted using a 4 by 3 contingency table. The degrees of freedom for this test equal this value.

Answer What is 6?

Worked Solution

$df = (\text{rows} - 1)(\text{columns} - 1) = (4 - 1)(3 - 1) = 3 \times 2 = 6.$

\$800 Before conducting a chi-square test, you must verify that all expected cell counts meet this minimum threshold.

Answer What is all expected counts must be at least 5?

Worked Solution

Large Counts condition for chi-square: all expected cell counts must be at least 5. If this fails, combine categories or collect more data.

\$1000 A researcher surveys 200 randomly selected people, recording political affiliation (Dem/Rep/Ind) and stance on a

policy (Support/Oppose). The null hypothesis for the appropriate chi-square test states this.

Answer What is: there is no association between political affiliation and policy stance in the population?

Worked Solution

This is a chi-square test of independence. H_0 : no association between political affiliation and policy stance. H_a : there is an association. $df = (3-1)(2-1) = 2$.